

Final Testing & Release Document

Historical v1.0.0 document — superseded by v2.0.0

This page consolidated the test artefacts for the **v1.0.0** public release and is retained for the audit trail. It has been superseded by the v2.0.0 implementation. Headline corrections to the claims below:

- The "proof generation" latencies (744.6 ms avg) were measured on a **SHA-256 digest simulation**, not SNARK execution. v2 removed the simulation: the SDK requires a real `MidnightProofProvider` and fails closed without one.
- The "29/30 (96.7%)" success table came from the **mock-adapter simulation harness**, not live rails.
- The five Preprod transactions ran against the **old signature-only validator**; the early REFUND succeeding demonstrates the **missing deadline enforcement**, and the RELEASE verified nothing beyond an operator signature.
- The Wise sandbox transfer `2147582543` was created but **never funded** (SCA-gated; final state `incoming_payment_waiting`).
- The Midnight deployment was a **placeholder-anchored local devnet run**, not bound to a real Cardano lock.

Current (v2.0.0) verification: Aiken validator **25/25** (`npm run cardano:check`) · Lucid emulator suite **17/17** (`sdk/test/escrow-emulator.test.mjs`) · backend API + oracle **15/15** (`npm run test:backend`) · `npm run typecheck` passes · E2E evidence in [docs/evidence/v2.0.0/](#) .

Item	Status
Tagged release	v1.0.0 — GitHub Release (superseded by the remediated v2.0.0 implementation on <code>main</code>)
Hosted developer docs	https://nucastio.github.io/MidnightZK-Off-Ramp-SDK-ADA-Web2-Payments/
Public GitHub repository	https://github.com/Nucastio/MidnightZK-Off-Ramp-SDK-ADA-Web2-Payments
Recorded demo walkthrough	docs/media/offramp-demo.mp4

1. Internal testing — v1.0.0 simulation harness (historical)

Label: these numbers were produced by the in-process **simulation harness** (`RAIL_ADAPTER_MODE=mock` , SHA-digest "proof" path). They are not live-rail or SNARK measurements. Real test counts for v2.0.0 are in the banner above.

A run of `npm run test:internal` against the v1.0.0 build reported **29 / 30 simulated successes (96.7%)**; the single failure was the mock adapters' intentional ~4-6% rejection injection exercising the `RailError` path.

Run timestamp: 2026-06-13T10:35:32.919Z **Total simulated off-ramps:** 30 (10 × cashapp / wise / revolut, all mock adapters) **Simulated success rate:** 96.7% **Avg simulated "prove" latency:** 744.6 ms — **this measured the SHA-256 simulation, not SNARK proving**, so it does not evidence the NFR-2 (≤ 50 s) budget for real proofs. Real Midnight proving times for v2 are recorded in the E2E evidence ([docs/evidence/v2.0.0/](#)).

Rail	Runs	Successes	Failures	Success rate	Avg "prove" (ms, simulated)	Avg submit (ms, mock)
cashapp	10	10	0	100.0%	742.69	0.27
wise	10	10	0	100.0%	763.34	1778.12
revolut	10	9	1	90.0%	727.76	0.17

2. Cardano Preprod — v1 validator (historical, superseded)

Five Preprod transactions were submitted against the **v1 signature-only validator** (script `addr_test1lrvzmkxhfm9j0u8g6p4cpkevqja4tn8qr88z7l7nc2tqrsxln25g`). They are listed with corrections in [testnet-evidence.md](#) . Contrary to the original wording, they did **not** exercise a "full validator surface": the v1 validator had no deadline, settlement, destination, or value checks — the early REFUND that was accepted is itself evidence of the missing deadline enforcement.

The v2 validator (oracle-signed UTxO-bound release authorization, deadline-gated refund, exact destinations, full value preservation) is verified by **25/25 Aiken tests** and the **17/17 Lucid emulator suite**, and on Preprod by the E2E runs in [docs/evidence/v2.0.0/](#) .

3. Wise sandbox — v1 evidence (historical; transfer unfunded)

The v1 Wise sandbox run created real provider objects (profile `30539072` , quote `fedc7ae5-...` , recipient `702406717` , transfer `2147582543`) but the transfer was **never funded**: the funding call returned 403 (SCA-gated) and the final observed state was `incoming_payment_waiting` . Raw captures remain under [docs/sandbox-evidence/](#) .

v2 status of the adapters:

- **Wise** — rewritten as a strict sandbox client: provider quote-bound transfer creation, deterministic idempotency, authenticated status polling, webhook signature verification, **no mock fallback**. Fresh evidence requires a fresh `WISE_API_TOKEN` (pending).

- **Revolut** — live sandbox **verified**: a real sandbox payment was completed through the adapter (refresh-token OAuth grant; JWT `iss` = certificate redirect-URI domain).
- **Cash App** — implemented against the **official Cash App Payouts API**; this is an early-access partner product and remains **credential-gated** — it must not be represented as live-evidenced.

4. Midnight ZK circuit — v1 local run (historical; placeholder-anchored)

The v1 local-devnet deployment (contract `a3a72a55...`, blocks 15768→15784) produced real SNARK proofs on a local node but was **placeholder-anchored** — not bound to any real Cardano lock. See the corrected record in [testnet-evidence.md](#). In v2, `bindOffRampIntent` anchors the real Cardano lock tx and receipts carry finalized Midnight tx/block/state data pinned by the 23-asset artifact manifest hash.

5. Backend API

`npm run dev` boots the Hono server (default port **8788**) serving the lifecycle as REST with a state machine, per-intent capability-token auth, adapter-observed settlement, and PII-redacting persistence — 15/15 tests. OpenAPI at `/docs` (Swagger) and `/api/openapi.json`. All routes documented in the [API reference](#). The v1 TryCloudflare "live MVP" tunnels are offline (they were ephemeral evaluation-window tunnels).

6. Issues fixed before the v1.0.0 release (historical)

Issue	Fix	Where
<code>libsodium-wrappers-sumo</code> ESM resolution	One-time shell fixup copying the file into the expected location (now run automatically on <code>npm install</code>).	scripts/fix-libsodium.sh
Mock-mode rail rejection wasn't reflected in the harness summary	Per-rail success-rate column + "Failures" section in the regenerated report.	scripts/internal-test.ts
Wise sandbox <code>/payments</code> SCA-gating	Documented as provider-side; correction : this left the v1 transfer unfunded, so the v1 Wise evidence shows an accepted-but-unfunded transfer, not a settled payout.	docs/sandbox-evidence/README.md

7. Acceptance-criteria checklist — corrected

Milestone criterion	v1.0.0 claim	Corrected status
Comprehensive developer docs published	✓	✓ — docs since rewritten to match the remediated implementation
Final testing + release document	✓	This page (historical) + the v2 banner above

Milestone criterion	v1.0.0 claim	Corrected status
Public GitHub repository	✓	✓
Recorded demo walkthrough video	✓	✓ (shows the v1 flow)
Tagged release	✓ v1.0.0	v1.0.0 superseded; remediated implementation on <code>main</code>
Transaction success rate $\geq 90\%$	"96.7%"	Simulation-harness figure (mock adapters). Real v2 counts: 25/25 Aiken, 17/17 emulator, 15/15 backend; E2E runs in <code>docs/evidence/v2.0.0/</code>
Proof generation $\leq 50\,000$ ms (NFR-2)	"744.6 ms avg"	Simulated (SHA digest, not SNARK). Real proving latency recorded in v2 E2E evidence
Smart contracts deploy and function on Cardano testnet	"5 Preprod txs"	v1 txs exercised a signature-only validator; the enforcing v2 validator is test-covered (25 + 17) with E2E Preprod evidence in <code>docs/evidence/v2.0.0/</code>
Real sandbox provider integration	"Wise transfer 2147582543 "	v1 Wise transfer was unfunded . v2: Revolut live sandbox verified (real payment completed); Wise strict client pending a fresh token; Cash App credential-gated
Midnight ZK circuit deployed	"4 SNARK proofs"	v1 run was local + placeholder-anchored; v2 receipts carry finalized, lock-anchored Midnight txs

8. Released artefacts

- **v1.0.0 GitHub Release** (superseded) — <https://github.com/Nucastio/MidnightZK-Off-Ramp-SDK-ADA-Web2-Payments/releases/tag/v1.0.0>
- **MIT LICENSE** — <https://github.com/Nucastio/MidnightZK-Off-Ramp-SDK-ADA-Web2-Payments/blob/main/LICENSE>
- **CHANGELOG** — <https://github.com/Nucastio/MidnightZK-Off-Ramp-SDK-ADA-Web2-Payments/blob/main/CHANGELOG.md>
- **Specifications:** [Project Initiation](#) · [SRS](#) · [TAD v1.1](#) (canonical)

9. Documentation link

Hosted developer documentation: <https://nucastio.github.io/MidnightZK-Off-Ramp-SDK-ADA-Web2-Payments/> — rebuilt automatically on every change to `docs/` or `mkdocs.yml` via `.github/workflows/docs.yml`. Start with the [trust model](#) for an honest statement of what each layer proves.